Smart-ASHA: Hardware Limitations & Environmental Post-Mortem

This document details the physical, electrical, and environmental challenges encountered during the breadboard prototyping of the Smart-ASHA diagnostic kit, including power constraints, thermal drift, and signal routing issues.

1. Power Rail Sag & Current Starvation

The Problem

During cold boot or concurrent sensor sampling, the ESP32-S3 intermittently reset with brownout notifications (`rst:0x10 (RTCWDT_RTC_RST)` or `rst:0xc (RTC_SW_CPU_RST)`). The OLED screen occasionally dimmed or locked during thermal camera acquisition.

Electrical Analysis

The total dynamic power budget of the prototype exceeded the current delivery limit of the onboard 3.3V Low-Dropout (LDO) regulator when powered via a standard laptop USB 2.0 port ($500\ \text{mA}$ limit):

- **ESP32-S3 (Dual-core active + vector processing):** Peak spikes up to $240\ \text{mA} - 310\ \text{mA}$.
- **MLX90640 Far-Infrared Sensor:** $23\ \text{mA}$ continuous, with internal capacitor charging transients.
- **MAX30105 Optical Sensor:** High-power LED pulse bursts ($0x3F \approx 12.5\ \text{mA}$ per LED at high sample rates).
- **AS7343 14-Channel Spectral Sensor:** $15\ \text{mA} - 20\ \text{mA}$.
- **SH1106 1.3-inch I2C OLED:** $25\ \text{mA} - 40\ \text{mA}$ depending on active pixel fill factor.

Physical Mitigation

1. **Capacitive Buffering:** Added a $100\ \mu\text{F}$ electrolytic capacitor in parallel with a $0.1\ \mu\text{F}$ ceramic capacitor directly across the primary 3.3V and GND rails of the solderless breadboard.
2. **Dedicated LDO Allocation:** Shifted high-drain optical emitters to dedicated voltage regulation, preventing digital voltage drops from resetting the sensitive I^2C sensor controllers.

2. Thermal Camera Radiometric Drift (Delhi Ambient Environment)

The Problem

In testing environments with high ambient room temperatures ($> 35^\circ\text{C}$ to 40°C), the MLX90640 raw pixel readings drifted significantly, reporting forehead temperatures above 44°C or falling below 32°C inside enclosed prototype housings.

Physical Root Causes

1. **Self-Heating of the Sensor Substrate:** The MLX90640 relies on thermopile elements measuring the temperature difference between the target object and the sensor silicon substrate ($V_{IR} \propto (T_{obj}^4 - T_{sub}^4)$). When placed in close proximity to the ESP32-S3 ($> 50^\circ\text{C}$ under AI load), heat transferred through jumper wires and shared air space raised T_{sub}, corrupting the internal gradient compensation.
2. **Enclosure Heat Trapping:** Without forced ventilation, heat from the display controller and microcontroller built up around the lens.

Engineering Compensation

- Implemented an experimental offset calibration factor (`boxOffset = -9.8 to -3.5` depending on enclosure draft).
- Implemented a **60-second High-Watermark Peak Extraction Algorithm**: Rather than relying on instantaneous single-frame reads, the system tracks the maximum valid pixel temperature over the duration of the scan, filtering out transient thermal blips.

3. Breadboard Parasitic Capacitance & I2C Bus Integrity

The Problem

Sensors intermittently failed communication with `NACK` errors, and the AS7343 repeatedly returned static `1` readings across all 18 channels.

Electrical Root Causes

1. **Bus Line Capacitance (C_b):** Connecting five separate I2C boards (TCA9548A, MLX90640, AS7343, MPU6050, SH1106) across long DuPont breadboard jumpers created a cumulative bus capacitance well in excess of the $400\ \text{pF}$ I^2C specification limit.
2. **Conflicting Onboard Pull-ups:** Each breakout module (Adafruit, SparkFun) included its own onboard $4.7\ \text{k}\Omega$ or $10\ \text{k}\Omega$ pull-up resistors to 3.3V. Operating in parallel through the multiplexer reduced the equivalent pull-up resistance (R_p) below $1\ \text{k}\Omega$, pulling excessive sink current and distorting logic low (V_{IL}) levels.

Physical Resolution

1. **Bus Segregation (Dual Physical Buses):** Moved the high-speed MAX30105 sensor completely off the shared bus onto an independent physical bus (`Wire1` : GPIO 16/17), freeing the multiplexer from handling continuous optical streaming.
2. **Frequency Reduction:** Reduced the primary I2C bus clock to $50\ \text{kHz}$ to allow the slow rise times caused by jumper wire capacitance to safely reach valid logic-high thresholds ($V_{IH} \geq 0.7 \times V_{DD}$).

4. Optical Isolation & Motion Artifacts

Limitations Identified in Prototype

1. **Spectral Sensor Optical Leakage:** The AS7343 requires controlled reflected light. On the open breadboard, ambient room lighting (fluorescent and LED flickering at $50\ \text{Hz}/100\ \text{Hz}$) overpowered the reflected optical signal from human tissue.
2. **MAX30105 Motion Sensitivity:** Solderless breadboard jumper flex during physical finger contact introduced movement artifacts into the PPG signal, corrupting pulse wave intervals.
3. **AD8232 ECG Noise Coupling:** The single-lead ECG signal path picked up significant $50\ \text{Hz}$ electromagnetic interference from nearby laptop chargers and USB power bricks, necessitating careful shielding and ADC filtering.

5. Transition to Production (Prototype vs. Final Sleeve)

Parameter	Breadboard Prototype (Current)	Production Stethoscope Sleeve (Target)
Interconnects	20cm jumper wires ($C_b > 300\ \text{pF}$)	4-layer rigid-flex PCB with controlled impedance
I2C Clock	Throttled to $50\ \text{kHz}$	Full $400\ \text{kHz}$ Fast-Mode
Thermal Isolation	Open air near hot ESP32-S3	Thermally isolated front baffle with silicone gasket
Optical Path	Open to ambient room light	Enclosed optical chamber with direct skin contact
Power Supply	USB VBUS through linear regulator	Dedicated low-noise LDO + LiPo battery management